

Description of files in this folder

1. run_trim_galore.sh
 - a. Command used to run TrimGalore on the samples
 - b. FASTQs are available under NCBI BioProject ID PRJNA1328687
2. run_kallisto.sh
 - a. Command used to run kallisto after trimming
3. sample_info_mouse_brain_rnaseq.xlsx
 - a. Metadata used for the analyses below
4. pca_and_deseq2.Rmd
 - a. Inputs:
 - i. The Kallisto output files which can be downloaded from GEO
 - ii. The t2g.txt that came with the standard mouse transcriptome index that was used by kallisto for pseudoalignment
 1. Downloaded from the following link on March 5, 2025:
 - a. https://github.com/pachterlab/kallisto-transcriptome-indices/releases/download/v1/mouse_index_standard.tar.xz
 - iii. sample_info_mouse_brain_rnaseq.xlsx (metadata, available in this folder, see above)
 - b. Outputs:
 - i. PCAs for each mouse brain RNA-seq experiment (there were two, see Methods) – used for a supplementary figure
 1. expt1_samples_pca.pdf
 2. expt2_samples_pca.pdf
 - ii. DESeq2 results, used to create a supplementary table
 1. Contained within deseq2_outputs subfolder
 - iii. Cook's distance results mentioned in the methods section (cooks_distance_tmao_samples.pdf)
 - iv. Volcano plots for each metabolite tested (TMAO, PA, IAA, GDCA) – used for a main figure
5. fgsea.Rmd
 - a. Inputs:
 - i. The DESeq2 results (produced by the previous script) for TMAO and GDCA
 - ii. The following collections of gene sets, downloaded from:
<https://www.gsea-msigdb.org/gsea/msigdb/mouse/collections.jsp>
 1. Hallmark gene sets
 2. REACTOME subset of CP (canonical pathways)
 - b. Outputs:

- i. All contained within the fgsea_results subfolder:
 - 1. Results of gene set enrichment analysis, using the input collections of gene sets
 - a. Used for a supplementary table
 - 2. PDFs depicting selected significant terms from GSEA
 - a. Used for figures
- 6. marker_gene_analysis_tmao.Rmd
 - a. Inputs:
 - i. Processed single-cell data from <https://doi.org/10.1038/s41593-019-0491-3>, downloaded from https://singlecell.broadinstitute.org/single_cell/study/SCP263/aging-mouse-brain
 - ii. Genes significantly regulated by TMAO, as outputted by the DESeq2 script above
 - b. Outputs:
 - i. Plot showing % of marker genes downregulated by TMAO, for each of six brain cell classes
 - 1. Used for a main figure
 - ii. RDS containing the computed marker genes per class
- 7. marker_gene_analysis_gdca.Rmd
 - a. Inputs:
 - i. The RDS outputted by the previous script
 - ii. Genes significantly regulated by GDCA, as outputted by the DESeq2 script above
 - b. Outputs:
 - i. Plot showing % of marker genes downregulated by GDCA, for each of six brain cell classes
 - 1. Used for a main figure
- 8. gaba_a_receptor_genes_heatmap.Rmd
 - a. Inputs:
 - i. Same as those for pca_and_deseq2.Rmd, above
 - b. Outputs:
 - i. A heatmap, used for a supplementary figure, showing downregulation of GABA_A receptor subunits in the brains of TMAO-treated mice
- 9. star_alignment_script.sh
 - a. Bash script that was used to run STAR alignment (instead of kallisto) on the trimmed fastq files, in order to prepare for differential splicing analysis with rMATS-turbo
- 10. star_index_script.sh

- a. Bash script that was used to create the STAR index for STAR alignment
11. rstats_turbo_script.sh
- a. Bash script that was used to run rMATS-turbo after STAR alignment
12. differential_splicing_events.Rmd
- a. Inputs:
 - i. JCEC files outputted by rMATS-turbo, available in this folder within the subfolder rstats_output
 - b. Outputs:
 - i. Excel files used for a supplementary table
 - ii. Volcano plot used for a main figure panel